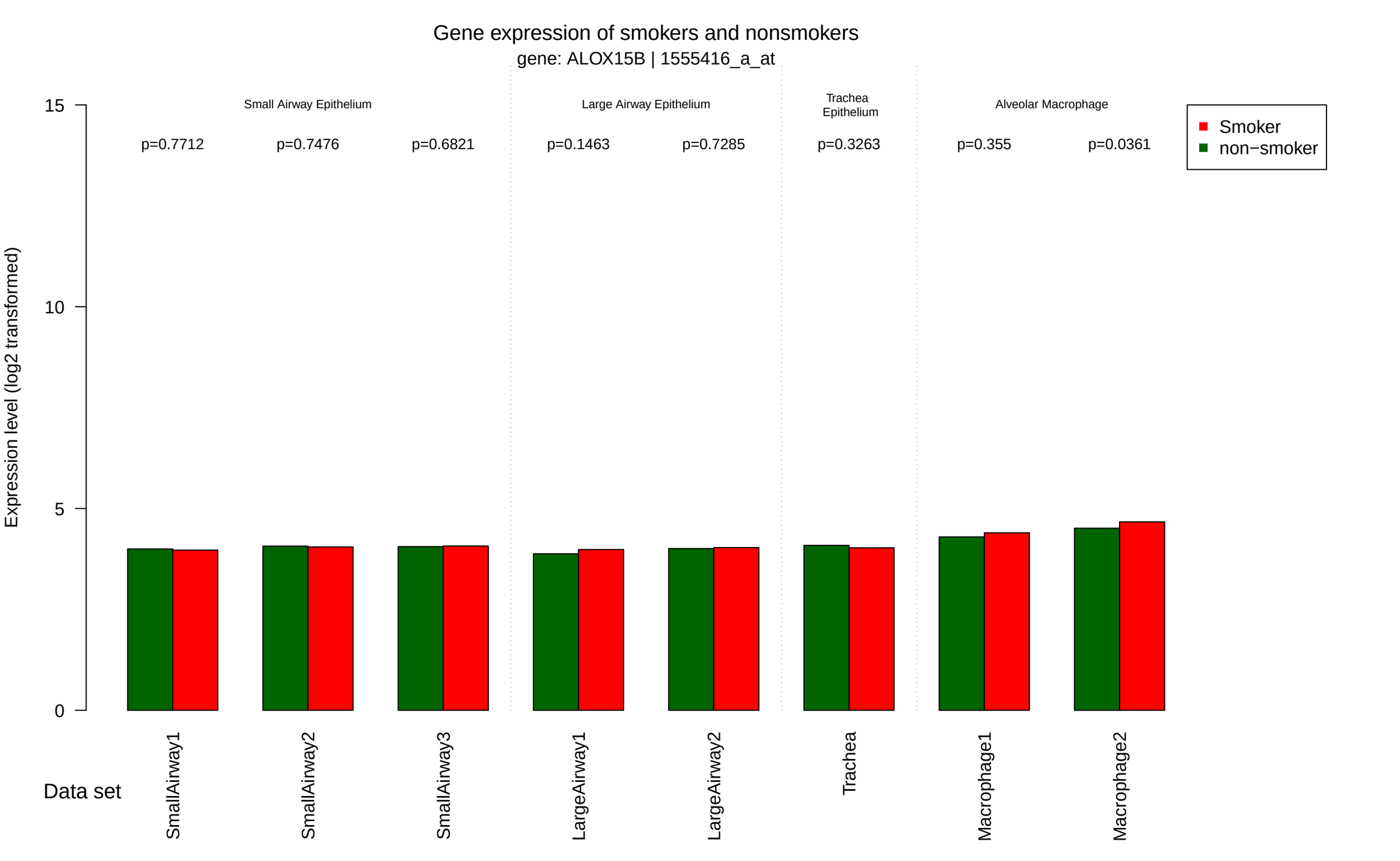
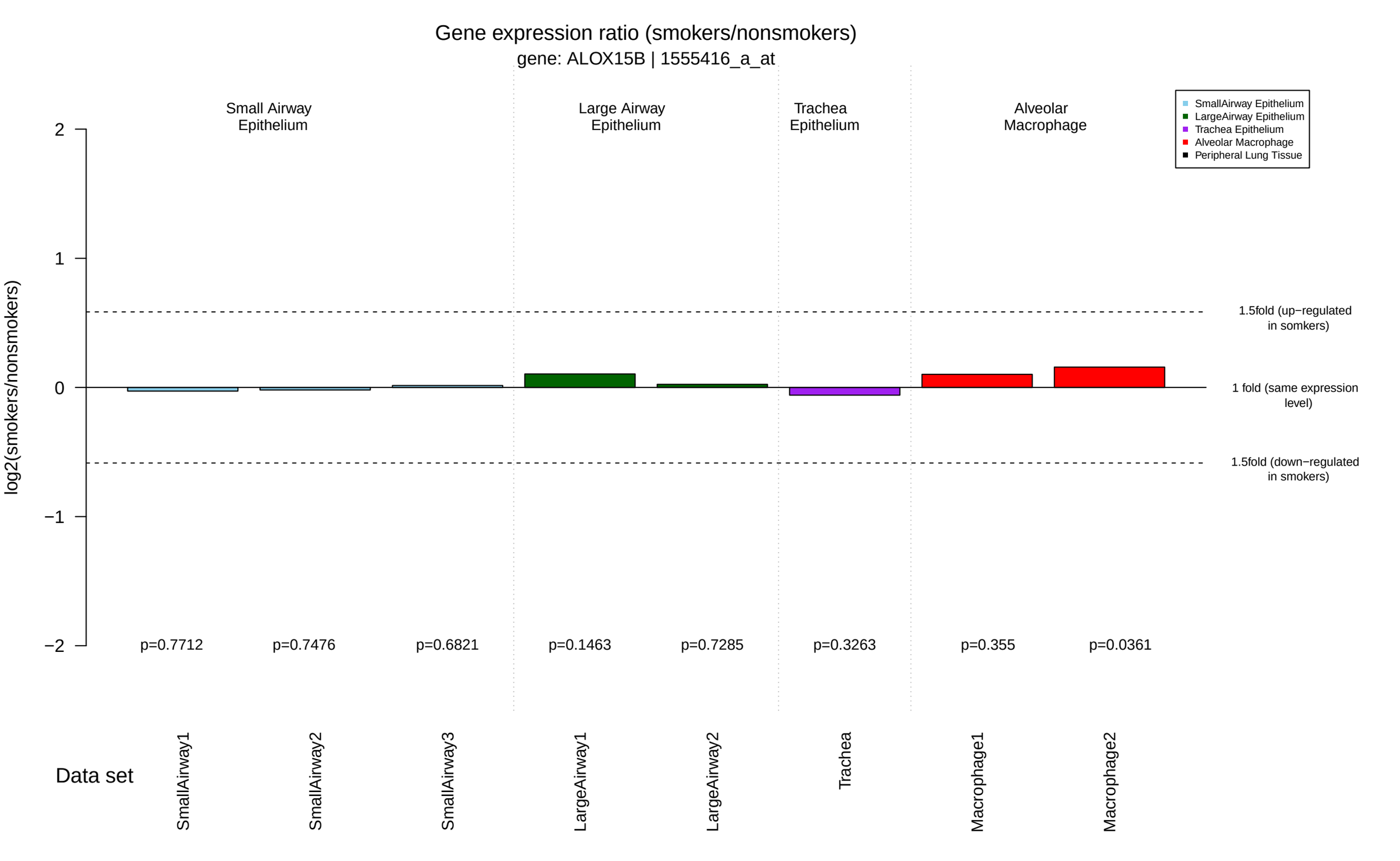
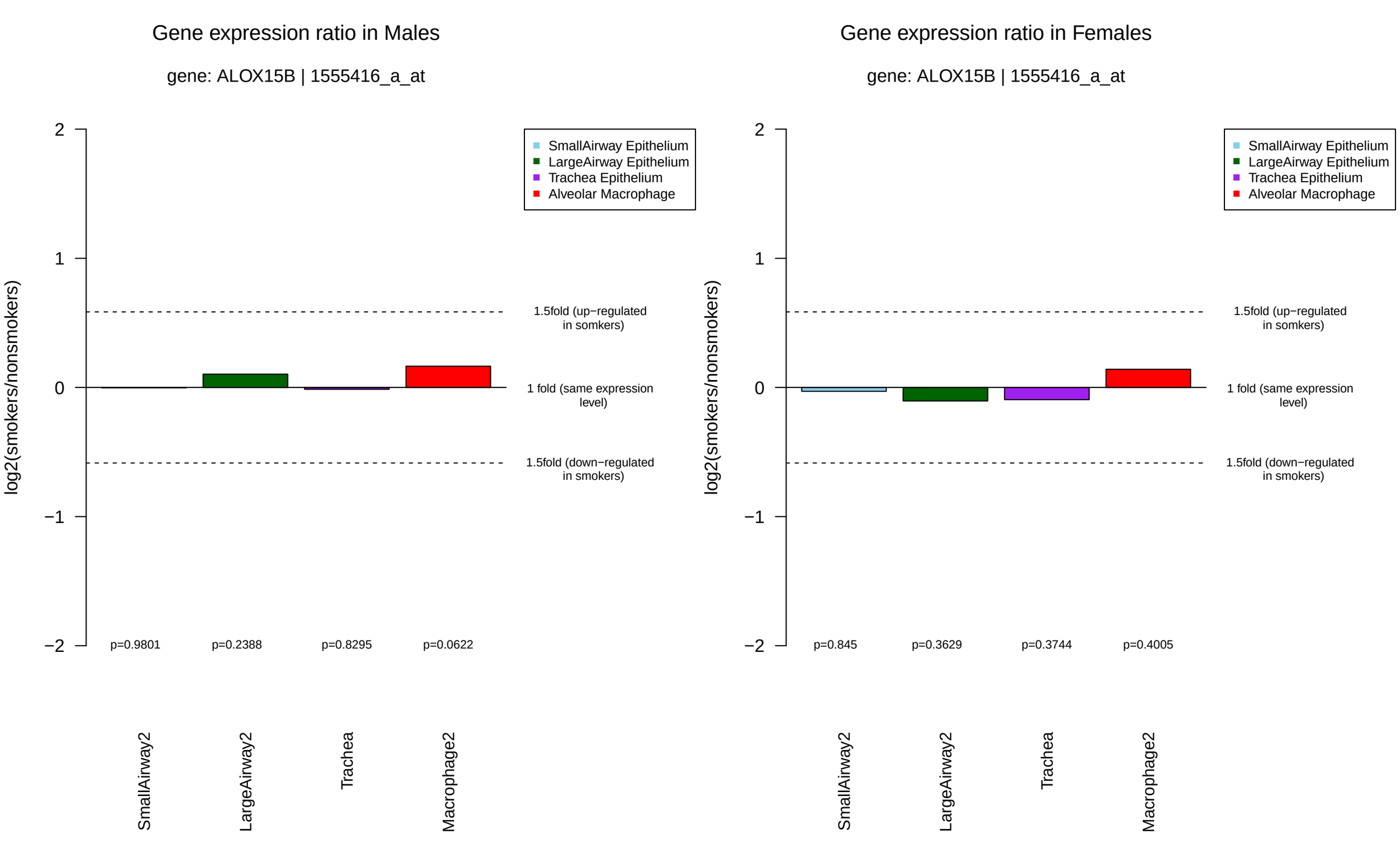
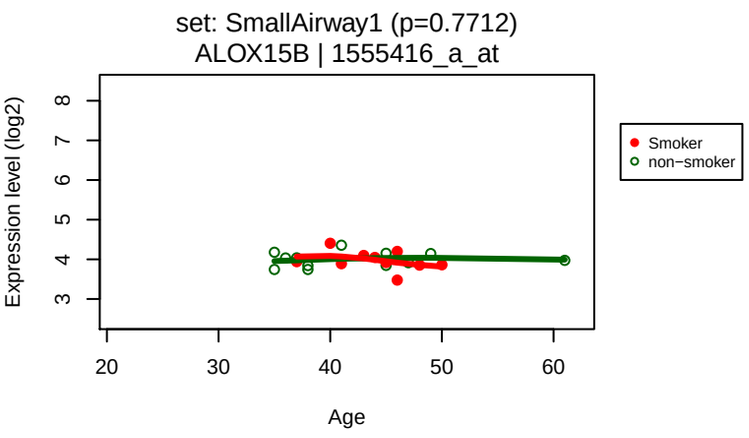
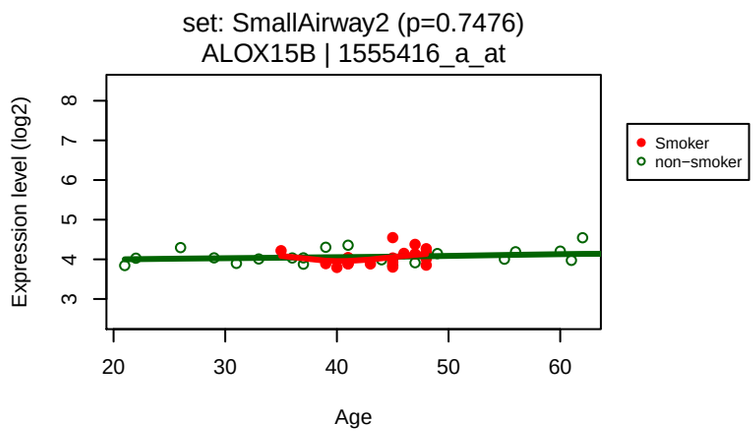
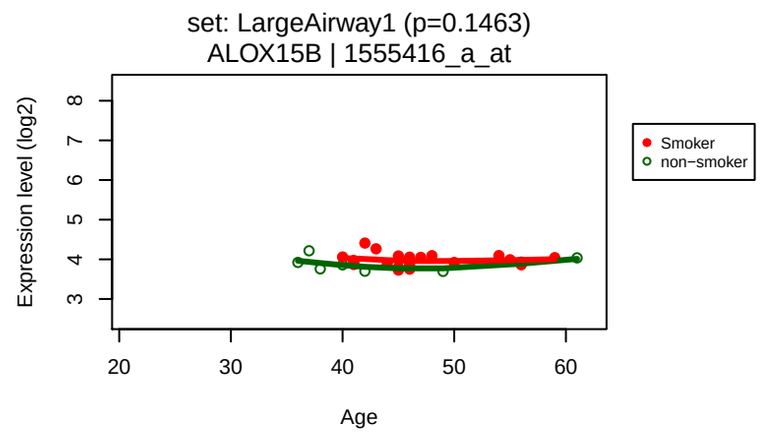

Gene expression of smokers and nonsmokers
gene: ALOX15B | 1555416_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7712
p=0.7476
p=0.6821
p=0.1463
p=0.7285
p=0.3263
p=0.355
p=0.0361
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

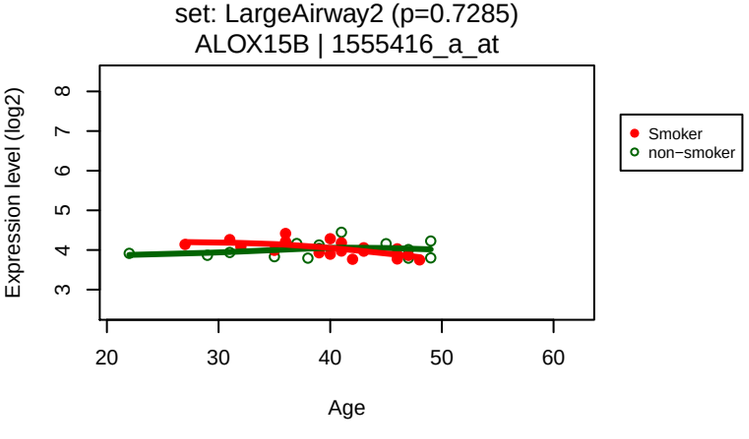

Gene expression ratio (smokers/nonsmokers)
gene: ALOX15B | 1555416_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7712
p=0.7476
p=0.6821
p=0.1463
p=0.7285
p=0.3263
p=0.355
p=0.0361
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

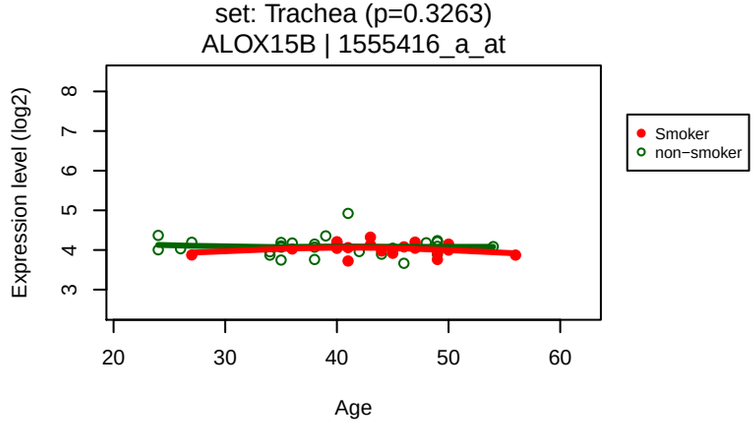

Gene expression ratio in Males
gene: ALOX15B | 1555416_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9801
p=0.2388
p=0.8295
p=0.0622
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALOX15B | 1555416_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.845
p=0.3629
p=0.3744
p=0.4005
SmallAirway2
LargeAirway2
Trachea
Macrophage2

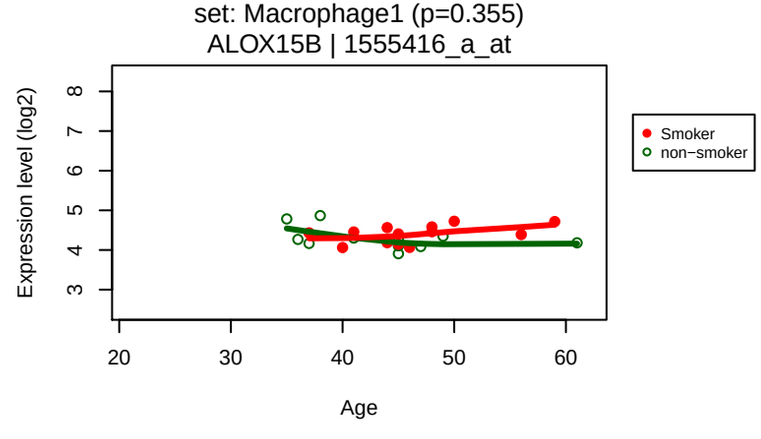

set: SmallAirway1 (p=0.7712)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker

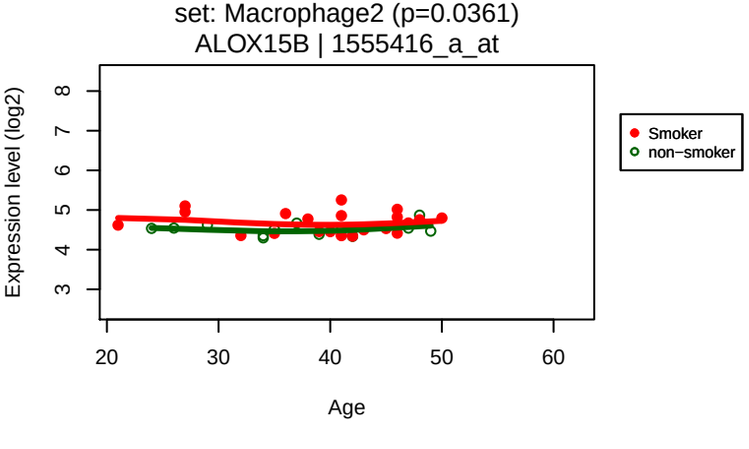

set: SmallAirway2 (p=0.7476)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker

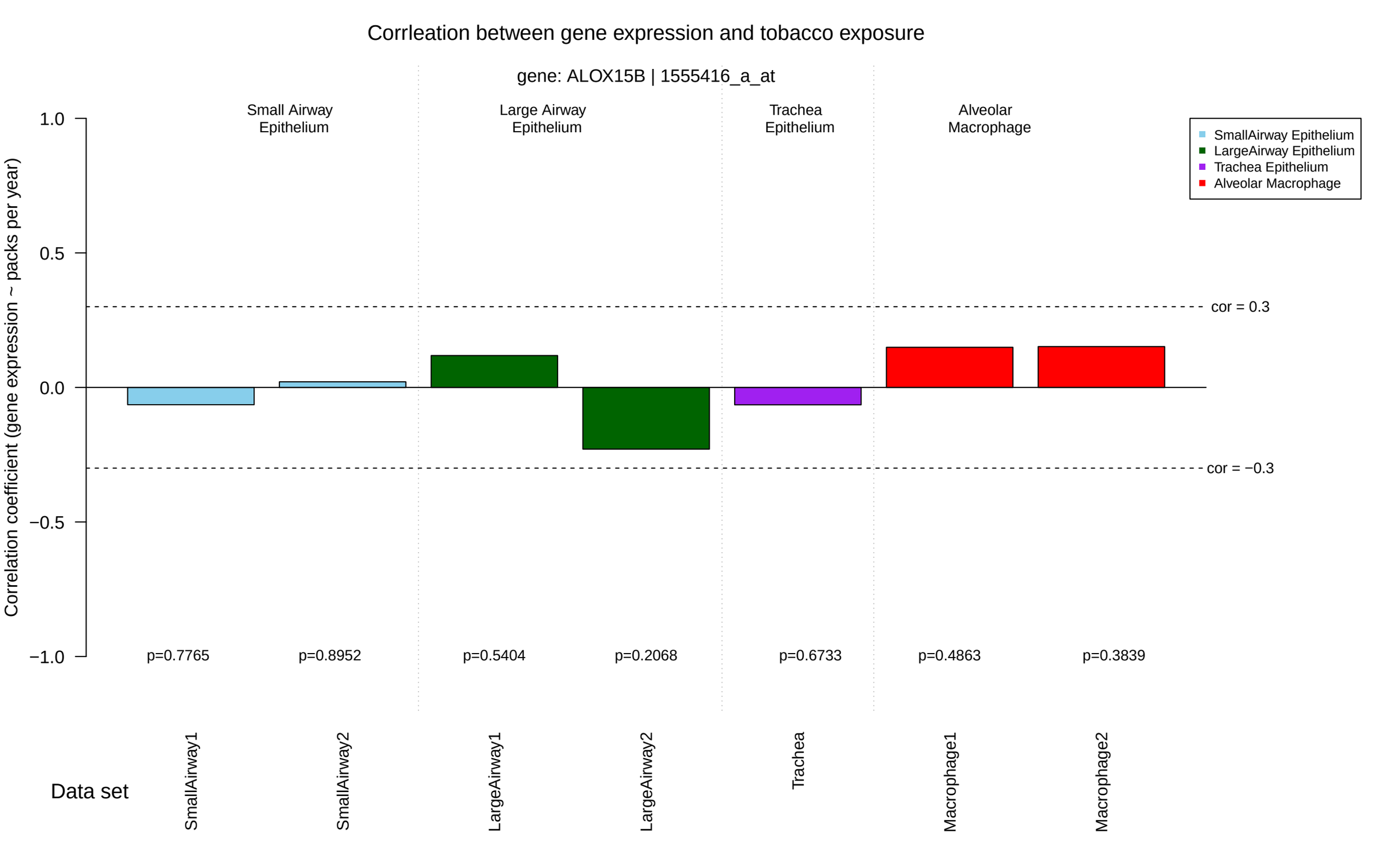

set: LargeAirway1 (p=0.1463)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7285)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3263)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.355)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0361)
ALOX15B | 1555416_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALOX15B | 1555416_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7765
p=0.8952
p=0.5404
p=0.2068
p=0.6733
p=0.4863
p=0.3839
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2